Lecture 20

Type A branching and Gelfand–Tsetlin patterns

Objectives. *We restrict the polynomial $\mathfrak{gl}_n$-modules of Lecture 18 to the upper-left block $\mathfrak{gl}_{n-1}$. Each summand occurs at most once. Repeating this along a chain of blocks splits a module into canonical lines. Restriction to $\mathfrak{sl}_{n-1}$ alone can repeat a summand.*

By the end of this lecture you will be able to:

- List the summands of $\mathbb{S}_\lambda(\mathbb{C}^n)$ restricted to $\mathfrak{gl}_{n-1}$ by interlacing.
- Prove this branching rule from the Schur interlacing identity and central scalars.
- Construct Gelfand–Tsetlin lines and state how far a basis is determined.
- Explain the two copies of V_1 in the restriction of the $\mathfrak{sl}_3$ adjoint module to $\mathfrak{sl}_2$.

Throughout, $n \geq 2$, all modules are finite-dimensional over $\mathbb{C}$, and E_{ij} is a matrix unit. For $1 \leq k \leq n$, let

$$\mathfrak{gl}_k = \operatorname{span}\{E_{ij} : 1 \leq i, j \leq k\} \subseteq \mathfrak{gl}_n, \qquad Z_k = E_{11} + \cdots + E_{kk}.$$

Here $\mathfrak{gl}_k$ is the upper-left $k \times k$ block. Its centre is $\mathbb{C}Z_k$, and

$$\mathfrak{gl}_k = \mathfrak{sl}_k \oplus \mathbb{C}Z_k, \qquad \mathfrak{sl}_1 = 0.$$

Let λ be a partition with at most n parts and size $d = |\lambda|$. Use the image model of Lecture 18 to realise the polynomial module $\mathbb{S}_\lambda(\mathbb{C}^n)$ as a submodule of $(\mathbb{C}^n)^{\otimes d}$. For a $\mathfrak{gl}_n$-module V and $a \in \mathbb{Z}^n$, the weight space is

$$V_a = \{v \in V : E_{ii}v = a_i v \text{ for } 1 \leq i \leq n\}.$$

If V is the sum of its weight spaces, its character is

$$\operatorname{ch} V = \sum_a \dim(V_a)\, x_1^{a_1} \cdots x_n^{a_n}.$$

By Lecture 18, $\operatorname{ch} \mathbb{S}_\lambda(\mathbb{C}^n) = s_\lambda(x_1, \ldots, x_n)$, the Schur polynomial.

20.1 The interlacing identity

Definition 20.1 (Interlacing)**.** Let $\lambda = (\lambda_1, \ldots, \lambda_n)$ be a partition, padded by zeros. An integer sequence $\mu = (\mu_1, \ldots, \mu_{n-1})$ *interlaces* λ, written $\mu \prec \lambda$, if

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \lambda_{n-1} \geq \mu_{n-1} \geq \lambda_n.$$

Such a μ is a partition with at most $n-1$ parts, since $\mu_{n-1} \geq \lambda_n \geq 0$.

Theorem 20.2 (Interlacing identity · *C*)**.** *For a partition λ with at most n parts,*

$$s_\lambda(x_1,\dots,x_n) = \sum_{\mu\prec\lambda} s_\mu(x_1,\dots,x_{n-1})\, x_n^{|\lambda|-|\mu|}. \tag{20.1}$$

Source: the finite-group course, Lecture 14 and its optional sheet. The tableau formula used below also appears in William Fulton and Joe Harris, Representation Theory: A First Course*, Exercise A.31(a).*

The tableau formula from the same lecture gives a direct derivation [C]:

$$s_\lambda(x_1,\dots,x_n) = \sum_T x^T.$$

The sum runs over semistandard tableaux T of shape λ with entries in $\{1,\dots,n\}$. The exponent of x_i in x^T is the number of entries i.

Delete the entries n from T. Rows weakly increase, so these entries end their rows. Columns strictly increase, so each column has at most one of them, at its bottom. What remains is a semistandard tableau T' of some shape $\mu\subseteq\lambda$ with entries at most $n-1$. Hence μ has at most $n-1$ rows.

The removed boxes lie in distinct columns. This holds exactly when

$$\mu_i \geq \lambda_{i+1} \quad \text{for } 1\leq i\leq n-1.$$

If $\mu_i<\lambda_{i+1}$, column μ_i+1 loses boxes in rows i and $i+1$. If rows $i<i'$ both lose a box in column j, then $\mu_i<j\leq\lambda_{i'}\leq\lambda_{i+1}$. Together with $\mu\subseteq\lambda$, the condition is $\mu\prec\lambda$.

Conversely, filling λ/μ with n turns any such T' into a valid T. The monomials satisfy

$$x^T = x^{T'}\, x_n^{|\lambda|-|\mu|}.$$

Summing over $\mu\prec\lambda$ and over T' of shape μ, and using the tableau formula in $n-1$ variables, gives (20.1).

Example 20.3 (Interlacing for $(2,1,0)$)**.** For $\lambda=(2,1,0)$ the conditions are $2\geq\mu_1\geq 1\geq\mu_2\geq 0$. Thus

$$\mu\in\{(2,1),\ (2,0),\ (1,1),\ (1,0)\},$$

and

$$s_{(2,1)}(x_1,x_2,x_3) = s_{(2,1)}(x_1,x_2) + \big(s_{(2)}(x_1,x_2)+s_{(1,1)}(x_1,x_2)\big)x_3 + s_{(1)}(x_1,x_2)\,x_3^2.$$

At $x_1=x_2=x_3=1$ this reads $8=2+(3+1)+2$.

20.2 Branching from $\mathfrak{gl}_n$ to $\mathfrak{gl}_{n-1}$

Lemma 20.4 (Weights and central scalars)**.** *Let $m\geq 1$, let λ have at most m parts, and let $V=\mathbb{S}_\lambda(\mathbb{C}^m)$.*

(a) V is the sum of its weight spaces. Every weight a has $a_i\in\mathbb{Z}_{\geq 0}$ and $a_1+\cdots+a_m=|\lambda|$.

(b) The central element $Z_m=I_m$ acts on V by the scalar $|\lambda|$.

Proof. For a standard basis tensor,

$$E_{jj}\,(e_{i_1}\otimes\cdots\otimes e_{i_d}) = \#\{r : i_r = j\}\; e_{i_1}\otimes\cdots\otimes e_{i_d}.$$

So the diagonal matrices act on $(\mathbb{C}^m)^{\otimes d}$ by commuting diagonalisable operators, with weights in $\mathbb{Z}^m_{\geq 0}$ of total $d = |\lambda|$. Commuting diagonalisable operators stay simultaneously diagonalisable on an invariant subspace, and the submodule V is invariant under them. This proves (a). Summing the entries of a weight gives (b). For $v \in V_a$,

$$Z_m v = (E_{11} + \cdots + E_{mm})v = (a_1 + \cdots + a_m)v = |\lambda| v. \qquad \blacksquare$$

Lemma 20.5 (Independence of Schur polynomials)**.** *Let $m \geq 1$. The polynomials $s_\mu(x_1, \ldots, x_m)$, for partitions μ with at most m parts, are linearly independent.*

Proof. For $m \geq 2$, this is the independence lemma of Lecture 19, proved by comparing the strictly decreasing monomials in the alternants. For $m = 1$, the Schur polynomials are $s_{(j)}(x_1) = x_1^j$ for integers $j \geq 0$. Distinct monomials are linearly independent. $\blacksquare$

Theorem 20.6 (Branching from $\mathfrak{gl}_n$ to $\mathfrak{gl}_{n-1} \cdot P$ given the interlacing identity)**.** *Let λ have at most n parts and $d = |\lambda|$. As a $\mathfrak{gl}_{n-1}$-module,*

$$\mathbb{S}_\lambda(\mathbb{C}^n) \cong \bigoplus_{\mu \prec \lambda} \mathbb{S}_\mu(\mathbb{C}^{n-1}),$$

each summand occurring once. The summand for μ is the $\mathfrak{gl}_{n-1}$-isotypic component of type μ, and E_{nn} acts on it by $d - |\mu|$.

Proof. Write $V = \mathbb{S}_\lambda(\mathbb{C}^n)$ and $Z = Z_{n-1}$.

Step 1: split by the centre of $\mathfrak{gl}_{n-1}$. By Lemma 20.4(a), Z is diagonalisable on V with eigenvalues in $\mathbb{Z}_{\geq 0}$:

$$V = \bigoplus_{k \geq 0} V[k], \qquad V[k] = \{v \in V : Zv = kv\}.$$

Since Z commutes with $\mathfrak{gl}_{n-1}$, each $V[k]$ is a $\mathfrak{gl}_{n-1}$-submodule. Lemma 20.4(b) gives

$$E_{nn} = I_n - Z = d - k \quad \text{on } V[k].$$

Step 2: irreducible pieces. By Weyl's theorem (Lecture 5), $V[k]$ is a direct sum of irreducible $\mathfrak{sl}_{n-1}$-modules U. Since Z acts on U by k, each U is an irreducible $\mathfrak{gl}_{n-1}$-submodule. For $n = 2$, take the U to be lines.

Step 3: identify each piece. The highest weight space of U is one-dimensional (Lecture 12). The diagonal of $\mathfrak{gl}_{n-1}$ is spanned by the diagonal of $\mathfrak{sl}_{n-1}$ and Z, so it preserves this space. Hence a highest weight vector u satisfies

$$E_{ii}u = \mu_i u \quad (1 \leq i \leq n-1), \qquad E_{nn}u = (d-k)u.$$

So u is a weight vector of V of weight $(\mu, d-k)$. Lemma 20.4(a), applied to this weight, gives

$$\mu_i \in \mathbb{Z}_{\geq 0}, \qquad \mu_1 + \cdots + \mu_{n-1} = d - (d-k) = k.$$

Dominance (Lecture 12) gives

$$\mu_i - \mu_{i+1} \in \mathbb{Z}_{\geq 0}, \qquad 1 \leq i \leq n-2.$$

So μ is a partition of k with at most $n-1$ parts. By Lecture 18, $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ is an irreducible $\mathfrak{sl}_{n-1}$-module with the same highest weight, and Z acts on it by $|\mu| = k$. By Lecture 13 there is an $\mathfrak{sl}_{n-1}$-isomorphism $U \cong \mathbb{S}_\mu(\mathbb{C}^{n-1})$. It commutes with Z, which is the scalar k on both sides, so it is a $\mathfrak{gl}_{n-1}$-isomorphism.

Step 4: count by characters. Steps 1–3 give multiplicities $m_\mu \geq 0$ with

$$V \cong \bigoplus_{\mu} \mathbb{S}_\mu(\mathbb{C}^{n-1})^{\oplus m_\mu},$$

and E_{nn} acts by $d - |\mu|$ on the copies of type μ. A vector of $\mathfrak{gl}_{n-1}$-weight b in such a copy has $\mathfrak{gl}_n$-weight $(b, d-|\mu|)$. Therefore

$$s_\lambda(x_1, \dots, x_n) = \sum_{\mu} m_\mu \, s_\mu(x_1, \dots, x_{n-1}) \, x_n^{d-|\mu|}.$$

Subtract (20.1). Put $\delta_\mu = 1$ if $\mu \prec \lambda$ and $\delta_\mu = 0$ otherwise. The coefficient of x_n^{d-k} gives

$$\sum_{|\mu|=k} (m_\mu - \delta_\mu)\, s_\mu(x_1, \dots, x_{n-1}) = 0 \quad \text{for each } k.$$

Lemma 20.5 gives $m_\mu = \delta_\mu$.

Step 5: isotypic components. Distinct μ give distinct $\mathfrak{gl}_{n-1}$-highest weights, so the summands are pairwise nonisomorphic. A type occurring once has that summand as its isotypic component (Lecture 14). ∎

20.3 Gelfand–Tsetlin patterns and lines

Apply the branching theorem along the chain

$$\mathfrak{gl}_1 \subset \mathfrak{gl}_2 \subset \cdots \subset \mathfrak{gl}_{n-1} \subset \mathfrak{gl}_n.$$

Definition 20.7 (Gelfand–Tsetlin pattern)**.** A *Gelfand–Tsetlin pattern* with top row λ is a sequence $\Lambda = (\lambda^{(n)}, \dots, \lambda^{(1)})$ with $\lambda^{(n)} = \lambda$ and

$$\lambda^{(k)} \in \mathbb{Z}^k \qquad (1 \leq k \leq n),$$
$$\lambda^{(k-1)} \prec \lambda^{(k)} \qquad (2 \leq k \leq n).$$

Each row $\lambda^{(k)}$ is a partition, by descending induction on k and the remark after the definition of interlacing. For $n = 3$ a pattern is written as a triangle, each entry between its upper neighbours:

$$\begin{array}{ccccc} \lambda_1 & & \lambda_2 & & \lambda_3 \\ & \lambda_1^{(2)} & & \lambda_2^{(2)} & \\ & & \lambda_1^{(1)} & & \end{array}$$

Fix $V = \mathbb{S}_\lambda(\mathbb{C}^n)$ and a pattern Λ. Define $V_\Lambda^{(n)} = V$. For $k < n$, let $V_\Lambda^{(k)}$ be the $\mathfrak{gl}_k$-isotypic component of type $\lambda^{(k)}$ in $V_\Lambda^{(k+1)}$. It depends only on rows $n, \dots, k$. Put $L_\Lambda = V_\Lambda^{(1)}$. A module isomorphism preserves every isotypic component, so it preserves these lines.

Theorem 20.8 (Gelfand–Tsetlin decomposition · *P*)**.** *(a)* $V_\Lambda^{(k)} \cong \mathbb{S}_{\lambda^{(k)}}(\mathbb{C}^k)$ *as* $\mathfrak{gl}_k$*-modules.*

(b) Each L_Λ *is a line, and* $V = \bigoplus_\Lambda L_\Lambda$ *over all patterns with top row* λ*.*

(c) E_{kk} *acts on* L_Λ *by* $|\lambda^{(k)}| - |\lambda^{(k-1)}|$, *where* $|\lambda^{(0)}| = 0$.
In particular, $\dim \mathbb{S}_\lambda(\mathbb{C}^n)$ *is the number of patterns with top row* λ.

Proof. (a) Use descending induction on k. The case $k = n$ is the definition. Suppose $V_\Lambda^{(k+1)} \cong \mathbb{S}_{\lambda^{(k+1)}}(\mathbb{C}^{k+1})$. A module isomorphism carries $\mathfrak{gl}_k$-isotypic components to $\mathfrak{gl}_k$-isotypic components. Theorem 20.6, with n replaced by $k+1$, therefore gives

$$V_\Lambda^{(k+1)} = \bigoplus_{\kappa \prec \lambda^{(k+1)}} V_\Lambda^{(k+1)}[\kappa], \qquad V_\Lambda^{(k+1)}[\kappa] \cong \mathbb{S}_\kappa(\mathbb{C}^k).$$

The component with $\kappa = \lambda^{(k)}$ is $V_\Lambda^{(k)}$.

(b) By (a) with $k = 1$, $L_\Lambda \cong \mathbb{S}_{\lambda^{(1)}}(\mathbb{C}^1)$, which is one-dimensional. The display in (a) splits $V_\Lambda^{(k+1)}$ over the possible rows k. Iterating from $k = n-1$ down to $k = 1$ gives $V = \bigoplus_\Lambda L_\Lambda$.

(c) The line L_Λ lies in $V_\Lambda^{(k)}$, on which Z_k acts by $|\lambda^{(k)}|$ by part (a) and Lemma 20.4(b). The same holds for $k-1$ when $k \geq 2$. For $k = 1$ put $Z_0 = 0$. Hence, for $v \in L_\Lambda$,

$$E_{kk}v = (Z_k - Z_{k-1})v = \big(|\lambda^{(k)}| - |\lambda^{(k-1)}|\big)v. \qquad \blacksquare$$

A *Gelfand–Tsetlin basis* is a choice of nonzero $\xi_\Lambda \in L_\Lambda$ for every pattern. The lines depend only on the chain of blocks, so they are canonical. The vectors are not canonical. Any other choice is

$$\xi'_\Lambda = c_\Lambda \xi_\Lambda, \qquad c_\Lambda \in \mathbb{C}^\times.$$

Write $\rho(x)$ for the operator by which $x \in \mathfrak{gl}_n$ acts on V, and $D(x)$ for its matrix in the basis (ξ_Λ). Then

$$\rho(x)\xi'_{\Lambda'} = c_{\Lambda'} \sum_\Lambda D(x)_{\Lambda\Lambda'}\xi_\Lambda = \sum_\Lambda \frac{c_{\Lambda'}}{c_\Lambda} D(x)_{\Lambda\Lambda'}\,\xi'_\Lambda.$$

So the diagonal entries of $D(x)$ do not change. By part (c) of Theorem 20.8, $D(E_{kk})$ is diagonal in every Gelfand–Tsetlin basis, with the same diagonal entries. Off-diagonal entries of $D(E_{k,k+1})$ depend on the normalisation. Exercise 20.5 proves that $E_{k,k+1}$ changes only row k of a pattern.

Distinct patterns may have the same row sums, hence the same weight. So the lines refine the weight spaces, as § 20.5 shows at the zero weight of $\mathfrak{sl}_3$.

20.4 Restriction to $\mathfrak{sl}_{n-1}$

Restriction to $\mathfrak{sl}_{n-1}$ forgets the action of Z_{n-1}.

Proposition 20.9 (Coincidences over $\mathfrak{sl}_{n-1}$ · *P*). *(a) Let* μ, μ' *be partitions with at most* $n-1$ *parts. Then* $\mathbb{S}_\mu(\mathbb{C}^{n-1}) \cong \mathbb{S}_{\mu'}(\mathbb{C}^{n-1})$ *as* $\mathfrak{sl}_{n-1}$*-modules if and only if* $\mu' - \mu = c(1,\dots,1)$ *for some* $c \in \mathbb{Z}$.

(b) Let λ *have at most* n *parts. The restriction of* $\mathbb{S}_\lambda(\mathbb{C}^n)$ *to* $\mathfrak{sl}_{n-1}$ *is multiplicity-free if and only if* $\lambda_i = \lambda_{i+1}$ *for some* $1 \leq i \leq n-1$.

Proof. (a) If $n = 2$, then $\mathfrak{sl}_1 = 0$. Both modules are one-dimensional, and every difference is a multiple of (1). Let $n \geq 3$. By Lecture 13, irreducible modules are isomorphic exactly when their highest weights agree. By Lecture 18, the Dynkin labels of $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ are $\mu_i - \mu_{i+1}$. The labels agree exactly when

$$\mu_i - \mu_{i+1} = \mu'_i - \mu'_{i+1} \quad (1 \leq i \leq n-2),$$

that is, when $\mu'_i - \mu_i$ does not depend on i.

(b) By Theorem 20.6 and (a), the multiplicity of a class is the number of $\mu \prec \lambda$ in one coset $\mu + \mathbb{Z}(1,\dots,1)$. Suppose μ and $\mu + c(1,\dots,1)$ both interlace λ, with $c \geq 1$. For every $i \leq n-1$,

$$\lambda_{i+1} \leq \mu_i < \mu_i + c \leq \lambda_i.$$

So a repeated class forces $\lambda_i > \lambda_{i+1}$ for all $i \leq n-1$. Conversely, if all these inequalities hold, then

$$(\lambda_2,\dots,\lambda_n) \quad \text{and} \quad (\lambda_2+1,\dots,\lambda_n+1)$$

both interlace λ and differ by $(1,\dots,1)$. ∎

Over $\mathfrak{gl}_{n-1}$ each isotypic component is irreducible, hence a canonical subspace. Over $\mathfrak{sl}_{n-1}$ a repeated class has a multiplicity space of dimension at least two. Its isotypic component splits into irreducibles in infinitely many ways, and the action of $\mathfrak{sl}_{n-1}$ distinguishes none of them. The operator Z_{n-1} commutes with $\mathfrak{sl}_{n-1}$ and selects one splitting, since it acts by different scalars on the members of one class:

$$|\mu + c(1,\dots,1)| = |\mu| + c(n-1).$$

Inside $\mathfrak{sl}_n$, the traceless element

$$t_n = Z_{n-1} - (n-1)E_{nn}$$

commutes with $\mathfrak{sl}_{n-1}$. By Theorem 20.6 it acts on the μ-summand by

$$|\mu| - (n-1)(d - |\mu|) = n|\mu| - (n-1)d.$$

So t_n separates members of one class. Hence the $\mathfrak{sl}_n$-module $\mathbb{S}_\lambda(\mathbb{C}^n)|_{\mathfrak{sl}_n}$ restricts multiplicity-freely to $\mathfrak{sl}_{n-1} \oplus \mathbb{C}t_n \cong \mathfrak{gl}_{n-1}$.

20.5 The $\mathfrak{sl}_3$ adjoint module restricted to $\mathfrak{sl}_2$

Take $n = 3$ and the standard $\mathfrak{sl}_2$ in the upper-left block:

$$h = E_{11} - E_{22}, \qquad e = E_{12}, \qquad f = E_{21}.$$

By Lecture 18, the adjoint $\mathfrak{sl}_3$-module is the restriction of $\mathbb{S}_{(2,1)}(\mathbb{C}^3)$.

Step 1: $\mathfrak{gl}_2$ labels. Example 20.3 and Theorem 20.6 give

$$\mathbb{S}_{(2,1)}(\mathbb{C}^3) \cong \mathbb{S}_{(2,1)}(\mathbb{C}^2) \oplus \mathbb{S}_{(2)}(\mathbb{C}^2) \oplus \mathbb{S}_{(1,1)}(\mathbb{C}^2) \oplus \mathbb{S}_{(1)}(\mathbb{C}^2),$$

in the order $\mu = (2,1), (2,0), (1,1), (1,0)$. Here

$$\begin{aligned} \dim &: \quad 8 = 2+3+1+2, \\ E_{33} = 3 - |\mu| &: \quad 0,\ 1,\ 1,\ 2. \end{aligned}$$

The four labels μ differ, so this splitting is canonical.

Step 2: the adjoint action of $\mathfrak{gl}_3$. Let $\mathfrak{gl}_3$ act on $\mathfrak{sl}_3$ by ad. The action is nontrivial, but I_3 acts by 0, so this module is not polynomial. Let $\mathbb{C}_{-\operatorname{tr}}$ be the line on which x acts by $-\operatorname{tr} x$. Then

$$\mathfrak{sl}_3 \cong \mathbb{S}_{(2,1)}(\mathbb{C}^3) \otimes \mathbb{C}_{-\operatorname{tr}} \quad \text{as } \mathfrak{gl}_3\text{-modules}.$$

Both sides restrict to the same irreducible $\mathfrak{sl}_3$-module. The identity acts on both by zero:

$$\mathrm{ad}(I_3) = 0, \qquad |(2,1)| - \operatorname{tr} I_3 = 3 - 3 = 0.$$

Hence an $\mathfrak{sl}_3$-isomorphism (Lecture 13) commutes with $\mathfrak{gl}_3 = \mathfrak{sl}_3 \oplus \mathbb{C}I_3$. On $\mathfrak{gl}_2$, the twist shifts every label by $(-1,-1)$ and every E_{33}-eigenvalue by -1. The twist changes no subspace, so it keeps the isotypic components and the Gelfand–Tsetlin lines.

Step 3: explicit summands. Put $t = E_{11} + E_{22} - 2E_{33}$. For $A \in \mathfrak{gl}_2$ and $i \le 2$, matrix multiplication gives

$$[A, E_{i3}] = \sum_{j=1}^{2} A_{ji} E_{j3}, \qquad [A, E_{3i}] = -\sum_{j=1}^{2} A_{ij} E_{3j}.$$

So $\operatorname{span}\{E_{13}, E_{23}\}$ is the $\mathfrak{gl}_2$-module $\mathbb{C}^2$, with highest weight vector E_{13} of weight $(1,0)$. Also $\operatorname{span}\{E_{31}, E_{32}\}$ is its dual, with highest weight vector E_{32} of weight $(0,-1)$. The block $\mathfrak{sl}_2$ is an ideal of $\mathfrak{gl}_2$, with highest weight vector E_{12} of weight $(1,-1)$. Finally, t commutes with $\mathfrak{gl}_2$. These four spans have distinct types and dimensions $2+3+1+2 = 8$, so they are the isotypic components of Step 1.

μ	$\mathfrak{gl}_2$ label	$\mathrm{ad}\, E_{33}$	Gelfand–Tsetlin basis	$\mathfrak{sl}_2$
$(2,1)$	$(1,0)$	-1	$E_{13},\ E_{23}$	V_1
$(2,0)$	$(1,-1)$	0	$E_{12},\ h,\ E_{21}$	V_2
$(1,1)$	$(0,0)$	0	t	V_0
$(1,0)$	$(0,-1)$	1	$E_{32},\ E_{31}$	V_1

Step 4: forget the centre. The $\mathfrak{sl}_2$-label is $\mu_1 - \mu_2$, so

$$\mathfrak{sl}_3 \cong V_2 \oplus V_1 \oplus V_1 \oplus V_0, \qquad 8 = 3 + 2 + 2 + 1.$$

The two copies of V_1 have $\mathfrak{gl}_2$ labels $(1,0)$ and $(0,-1)$, which differ by $(1,1)$. Over $\mathfrak{sl}_2$ they are isomorphic. Indeed,

$$\begin{aligned} \mathrm{ad}(f)E_{13} &= E_{23}, \qquad \mathrm{ad}(f)E_{32} = -E_{31}, \\ \mathrm{ad}(e)E_{23} &= E_{13}, \quad \mathrm{ad}(e)(-E_{31}) = E_{32}, \end{aligned}$$

and $\mathrm{ad}(e)$ maps E_{13} and E_{32} to 0. Define the linear map $T\colon\ \operatorname{span}\{E_{13}, E_{23}\} \to \operatorname{span}\{E_{31}, E_{32}\}$ by

$$T(E_{13}) = E_{32}, \qquad T(E_{23}) = -E_{31}.$$

It matches highest weight vectors and their f-images in two copies of V_1, so it is an $\mathfrak{sl}_2$-isomorphism. It is not a $\mathfrak{gl}_2$-isomorphism, because Z_2 acts by different scalars:

$$\mathrm{ad}(Z_2)E_{13} = E_{13}, \qquad \mathrm{ad}(Z_2)E_{32} = -E_{32}.$$

The vectors of h-weight 1 that $\mathrm{ad}(e)$ maps to 0 form $\operatorname{span}\{E_{13}, E_{32}\}$. Every nonzero vector there generates a copy of V_1. For example, $E_{13} + E_{32}$ generates $\operatorname{span}\{E_{13} + E_{32},\ E_{23} - E_{31}\}$. Inside $\mathfrak{sl}_3$, $t = t_3$ separates the copies:

$$\mathrm{ad}(t)E_{13} = 3E_{13}, \qquad \mathrm{ad}(t)E_{32} = -3E_{32},$$

in agreement with $3|\mu| - 6$ for $|\mu| = 3$ and $|\mu| = 1$. Since $2 > 1 > 0$, Proposition 20.9 predicts this repetition.

Step 5: Gelfand–Tsetlin lines. The bottom entry ν satisfies $\mu_1 \geq \nu \geq \mu_2$, giving 2+3+1+2 = 8 patterns. In each block the E_{11}-eigenvalues are $\nu - 1$, hence distinct. So the lines are spanned by the table entries, listed with ν decreasing. The zero weight space has the two patterns with $|\mu| = 2$ and $\nu = 1$. It splits into the two lines

$$\mathbb{C}h \oplus \mathbb{C}t.$$

The second line is $\mathbb{C}t$, not $\mathbb{C}(E_{22} - E_{33})$. The matrix entries still depend on the scaling. For example,

$$\mathrm{ad}(e)E_{21} = h = \frac{1}{2}(2h).$$

After replacing h by $2h$, this matrix entry becomes $\frac{1}{2}$.

Self-checks

(1) How many Gelfand–Tsetlin patterns have top row $(1, 1, 0)$?
(2) Restrict $\mathbb{S}_{(2,2)}(\mathbb{C}^3)$ to $\mathfrak{sl}_2$. Is the result multiplicity-free?
(3) After $\xi_\Lambda \mapsto c_\Lambda \xi_\Lambda$, which product of entries of $D(E_{12})$ and $D(E_{21})$ is unchanged?

Answers. (1) Three: $\mu = (1, 1)$ with $\nu = 1$, and $\mu = (1, 0)$ with $\nu \in \{0, 1\}$. This is $\dim \Lambda^2\mathbb{C}^3$. (2) $\mu \in \{(2, 2), (2, 1), (2, 0)\}$ gives $V_0 \oplus V_1 \oplus V_2$, with $6 = 1 + 2 + 3$. It is multiplicity-free, as $\lambda_1 = \lambda_2$. (3) $D(E_{12})_{\Lambda\Lambda'} D(E_{21})_{\Lambda'\Lambda}$, since the factors $c_{\Lambda'}/c_\Lambda$ and $c_\Lambda/c_{\Lambda'}$ cancel.

Summary

Restricting $\mathbb{S}_\lambda(\mathbb{C}^n)$ to $\mathfrak{gl}_{n-1}$ gives each $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ with $\mu \prec \lambda$ exactly once. The proof combines the interlacing identity, complete reducibility and the central scalars $|\mu|$.

Iterating along $\mathfrak{gl}_1 \subset \cdots \subset \mathfrak{gl}_n$ gives canonical lines indexed by Gelfand–Tsetlin patterns. Basis vectors in them are fixed only up to independent scalars. Over $\mathfrak{sl}_{n-1}$, summands whose labels differ by full columns become isomorphic, as the two copies of V_1 in $\mathfrak{sl}_3$ show.

Reading

The finite-group course, Lecture 14 and its optional sheet, for the interlacing identity and the tableau formula. William Fulton and Joe Harris, *Representation Theory: A First Course*, §15. Roe Goodman and Nolan R. Wallach, *Symmetry, Representations, and Invariants*, Chapter 8, especially §8.1 on branching for general linear groups. Exercise Sheet 20 gives further calculations.